

INTRODUCTION TO SUSTAINABILITY



INTRODUCTION TO SUSTAINABLE DESIGN & CONSTRUCTION

Sustainable design seeks to reduce negative impacts on the environment, and the health and comfort of building occupants, thereby improving building performance. The basic objectives of sustainability are to reduce consumption of non-renewable resources, minimize waste, and create healthy, productive environments.

Sustainable design is known by some different names such as **ecological design, green design, green architecture, eco-effective and environmentally friendly design** which broadly encompasses a similar meaning.



PRINCIPLES OF SUSTAINABLE DESIGN/ CONSTRUCTION

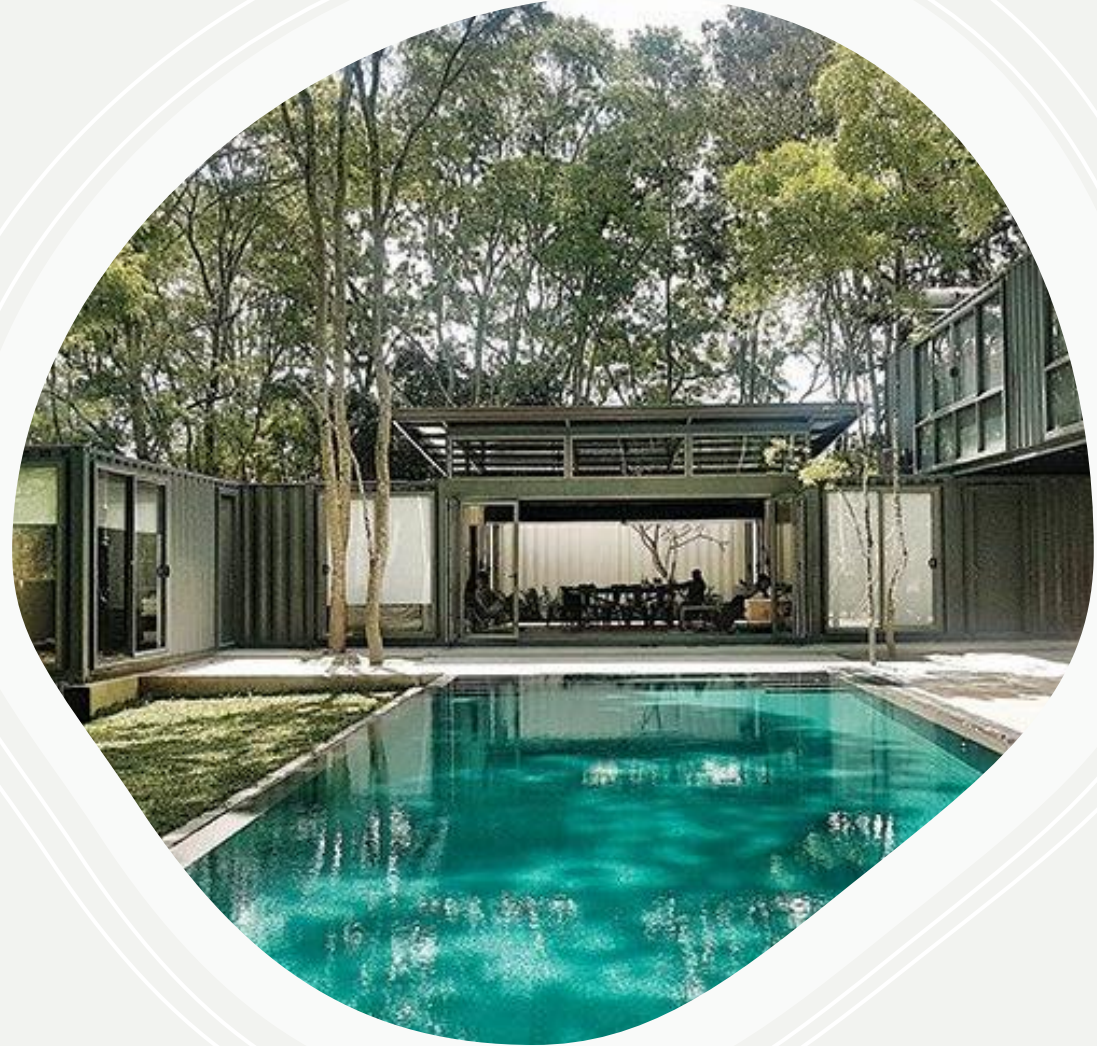
1. Reduce resource consumption (reduce)
2. Reuse resources (reuse)
3. Use recyclable resources (recycle)
4. Protect nature (nature)
5. Eliminate toxics (toxics)
6. Apply life-cycle costing (economics)
7. Focus on quality (quality)



SUSTAINABLE CONSTRUCTION – AREAS OF CONCERN

The issue of resource-conscious design is central to sustainable construction, which ultimately aims to minimize natural resource consumption and the resulting impact on ecological systems. Sustainable construction considers the role and potential interface with ecosystems to provide services in a synergistic fashion. With respect to materials selection, closing materials loops and eliminating solid, liquid, and gaseous emissions are key sustainability objectives.

Closed loop describes a process of keeping materials in productive use by reuse and recycling rather than disposing of them as waste at the end of the product or building life cycle. Products in closed loops are easily disassembled, and the constituent materials can be recycled and worthy of recycling.



AREA 1 – LAND USE

Sustainable land use is based on the principle that land, particularly undeveloped, natural, or agricultural land (greenfield), is a precious finite resource and its development should be minimized. Effective planning is essential for creating efficient urban forms and minimizing urban sprawl, which leads to overdependence on automobiles for transportation, excessive fossil fuel consumption, and higher pollution levels. Like other resources, land is recyclable and should be restored to productive use whenever possible.



AREA 2 - ENERGY AND ATMOSPHERE

Energy conservation is best addressed through effective building design, which integrates three general approaches:

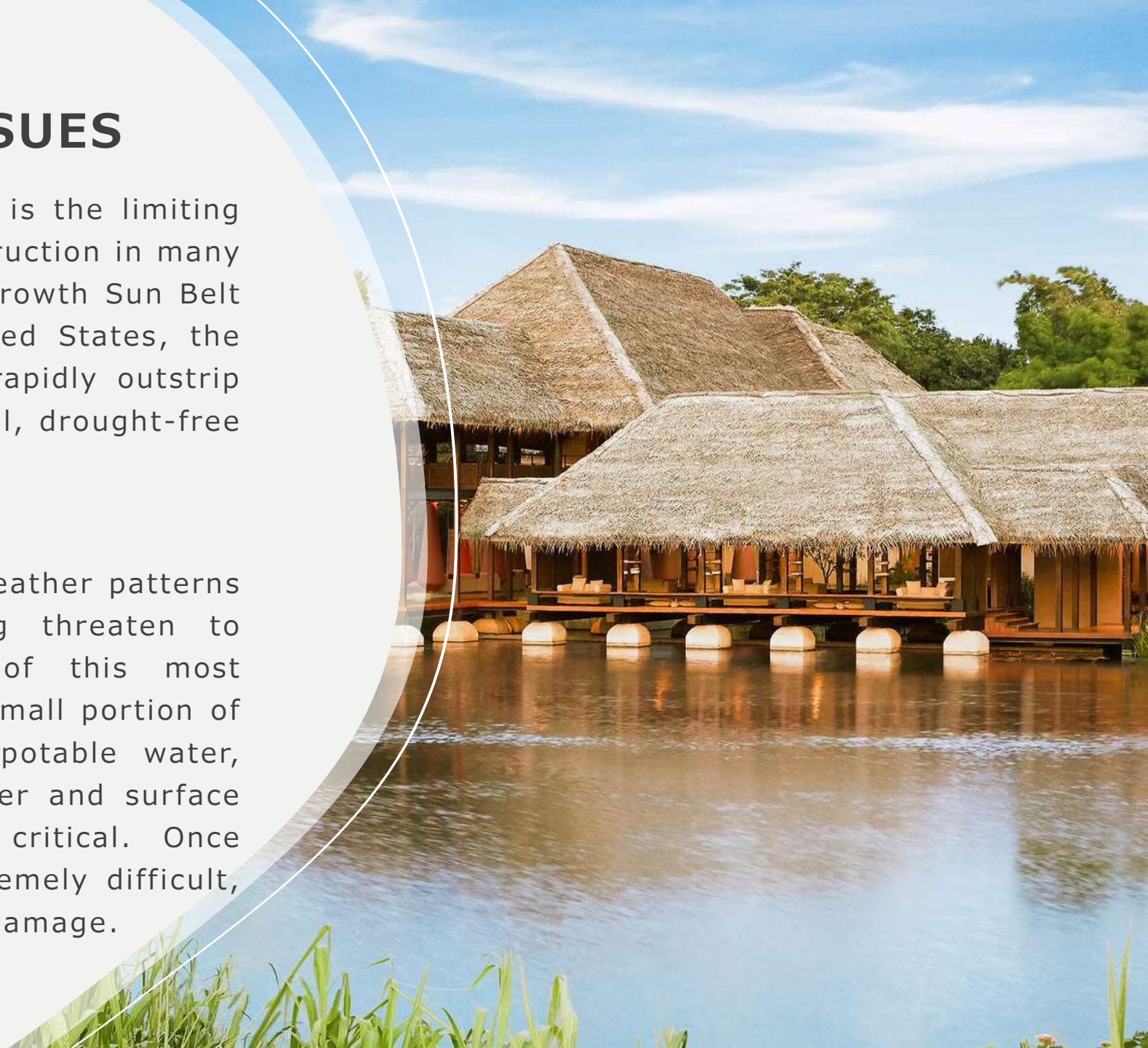
- (1) Fully implementing passive design
- (2) Designing a building envelope that is highly resistant to conductive, convective, and radiative heat transfer
- (3) Employing renewable energy resources



AREA 3 - WATER ISSUES

The availability of potable water is the limiting factor for development and construction in many areas of the world. In the high-growth Sun Belt and western regions of the United States, the demand for water threatens to rapidly outstrip the natural supply, even in normal, drought-free conditions.

Climate alterations and erratic weather patterns precipitated by global warming threaten to further limit the availability of this most precious resource. Since only a small portion of Earth's hydrologic cycle yields potable water, protection of existing groundwater and surface water supplies is increasingly critical. Once water is contaminated, it is extremely difficult, if not impossible, to reverse the damage.





SUSTAINABILITY AND COMMERCIAL VIABILITY

It has frequently been argued that sustainable design and commercial viability pull in opposite directions; that the capital cost of incorporating sustainability into a new development increases unit costs to such a level that it cannot compete in the marketplace.

Despite the significantly high initial costs, there are several financial and other benefits that a firm gains by incorporating sustainable design principles in building construction.

CASE STUDY

NSBM GREEN

UNIVERSITY DESIGN

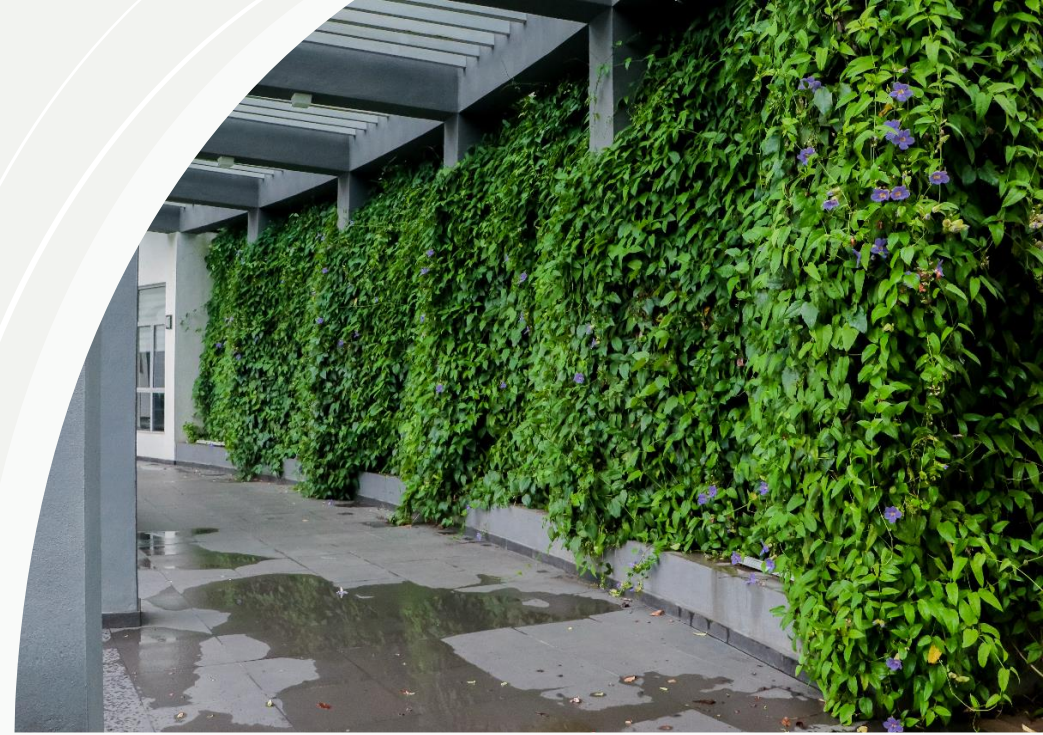
NSBM Green University is a purpose-built campus complex with a clearly defined location and facility with a strong underlying focus on sustainability and the environment. The University's overall architectural design harmonizes with the existing physical and natural environment of the area. The entire campus visually reconnects individual buildings ensuring continuous interaction between students and faculty.



GREEN WALLS

A green wall is defined as a system in which plants grow in a vertical surface such as a building façade in a controlled fashion and with regular maintenance.

In the case of NSBM Green University, the green walls assist the overall environment in **air purification, reduction of ambient temperature, reduction in external noise entering buildings while contributing to the overall beautification** of the University premises.



OPTIMAL USE OF NATURAL FLORA AND FAUNA

Natural flora has been optimally used across the university to further beautify the overall environment. Over 300 shade and flowering trees have been planted, which eventually gives a natural canopy with several environmental benefits.

These benefits include natural filtration of air, cooling of the overall climate and creation of a conducive climate for staff and students to engage in their work peacefully.



PHOTOGRAPHED BY : Chomod Nimantha



OPEN SPACE DESIGN

Large open spaces of double and triple heights supplemented with open courtyards is a common sighting across all the university buildings, especially in the main faculties. The university also houses many open and informal learning spaces where students can gather for study and group discussions.

This has allowed the University to capitalize on **natural ventilation** and **natural lighting** optimally across the entire university.



NATURAL ENVIRONMENT PRESERVATION

The existing natural terrain of the area has been retained with tremendous success offering an aesthetic appeal to the campus where natural boulders can be seen throughout the premises, thus contributing to a breathtaking backdrop while also preserving the original natural environment as it is.

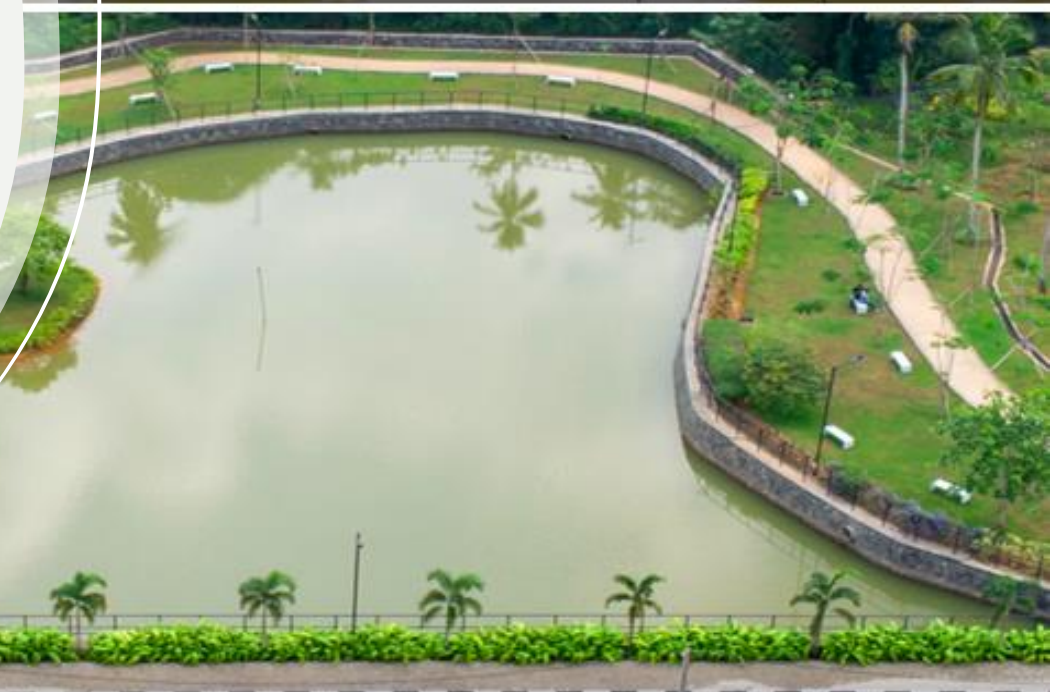


RAINWATER HARVESTING

Rainwater is primarily collected in 3 ponds located in Phase I and Phase II which are used to beautify and improve the natural ventilation of the university. Any excess water flows directly into the rainwater collecting lake through the university's drain system. Additionally, the rainwater accumulated near the areas of the auditorium building, main entrance, and student center building area's goes directly to the rainwater collecting lake through the drain system. The rainwater collected on the rooftop of other buildings is directed to the external canal through the pipes and drain system



lecting lake, NSBM Green University - Phase I



Rainwater collecting lake, NSBM Green University - Phase II

SUPPLEMENTING SUSTAINABLE DESIGN PRINCIPLES REDUCING PLASTIC POLLUTION

The University has installed water filters in key public and staff areas for both staff and students to refill water bottles as and when needed. Academic staff members have been presented with refillable glass water bottles for personal use during office hours. Students too are strongly encouraged to use re-fillable water bottles when they come to the University.



SUPPLEMENTING SUSTAINABLE DESIGN PRINCIPLES

COMPOSTING

The University uses several landscaping-based wastes generated (primarily grass) to produce compost within the university premises itself. The compost is produced in an open pit and the produced manure is re-used internally for gardening purposes within the University's gardens again.



SUPPLEMENTING SUSTAINABLE DESIGN PRINCIPLES

TREE PLANTING & GIFTING

NSBM Green University has also adopted a practice of **planting trees to mark special milestones** of the University across a wide variety of important dates and occasions including visits by VIP guests to the University. NSBM Green University has also commenced **gifting plants and plant saplings** as tokens of appreciation and gifts for special guests attending the University's events in order to broaden its environmental responsibility beyond the internal environment of the University.

